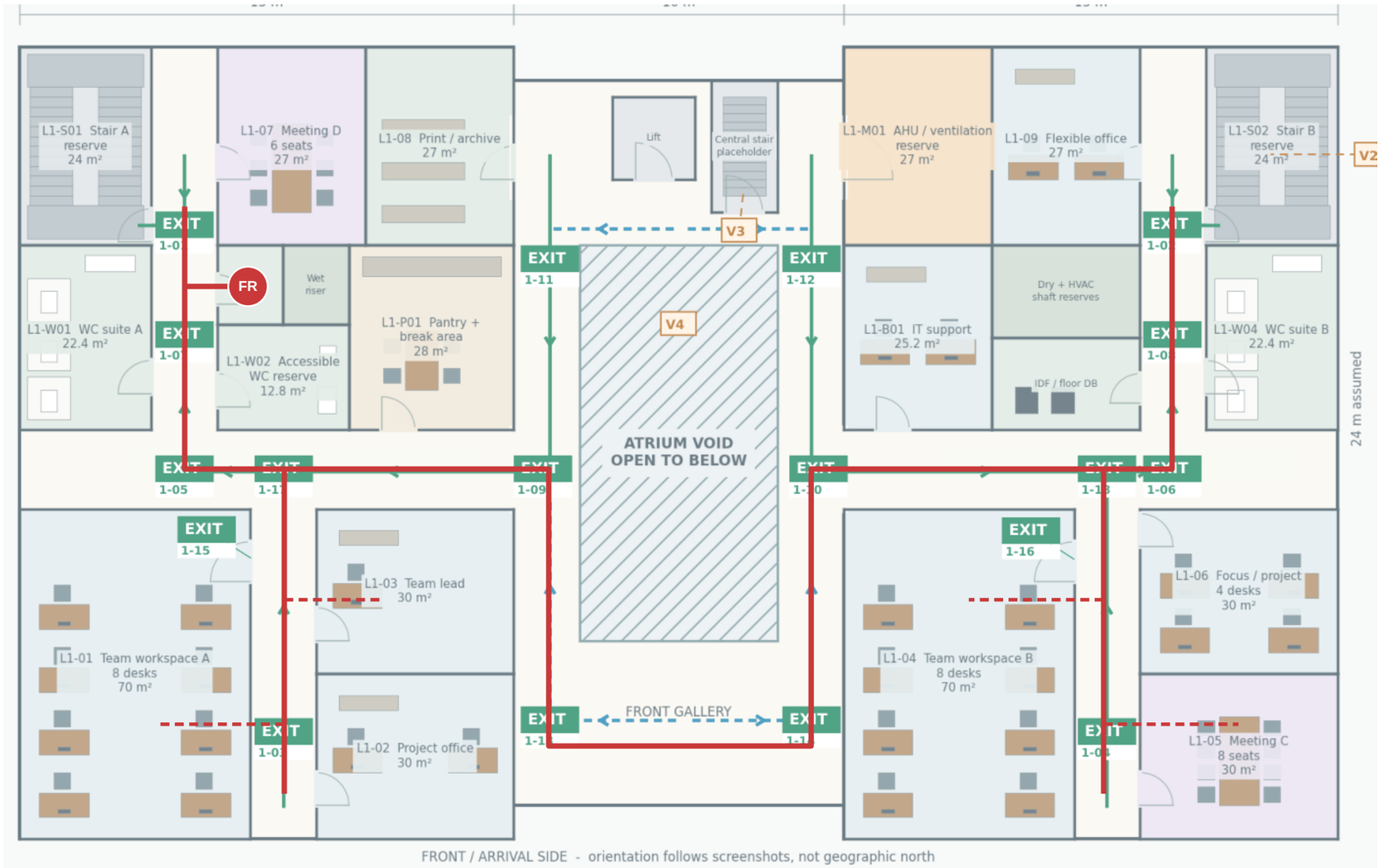
Fire water | Reserve a dedicated riser

First floor shown as a routing study. Use the same corridor logic on other floors, adjusted to their ceilings, compartments and actual geometry.

RED: candidate floor mains. **Dashed red:** example room branches only, not sprinkler coverage. **FR:** proposed new fire-riser reservation beside the left wet-service zone; space has not been established.



Start at ground level

Supply / proposed tank -> G-M01 fire pump room, if a pump is required -> dedicated fire distribution riser. The main route begins at this equipment, not at the domestic booster.

Reserve the vertical route

FR is a proposed location near the janitor/wet-service zone. It would take coordinated space on every floor. Do not assume the existing small plumbing shaft has spare capacity.

Branch on each floor

Reserve an accessible floor control assembly: isolation and supervision, waterflow indication, pressure gauge and test/drain provisions as required by the final design.

Connect the wings

A candidate crossing follows the front gallery perimeter, as drawn. Its ceiling/support space is unverified. If it cannot fit, assess another supported route or a separately fed riser.

Add detail later

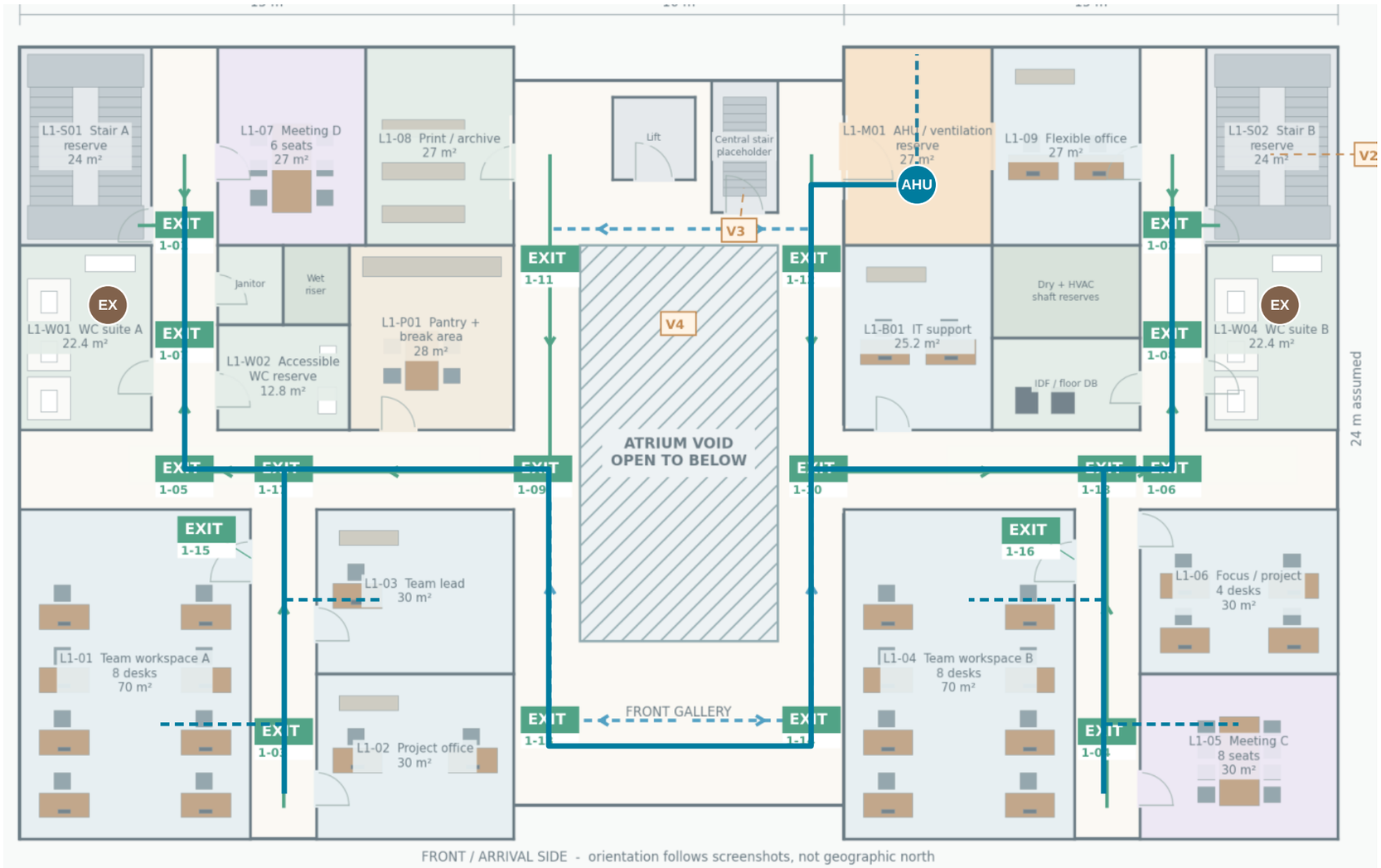
Route room branches after the floor mains are coordinated. Sprinkler positions, pipe diameters, pump duties, standpipes and fire-department connections require the fire-protection design.

Model now: FR-01, a floor-control reservation on G/L1/L2, corridor main centerlines, test/drain route reservations and example branches. Keep exits and stair enclosures available. **Do not model the red route as a validated hydraulic arrangement.**

HVAC | Separate air, refrigerant and drainage

The teal route shows a candidate ventilation supply network from L1-M01. It is not a return-air or toilet-extract route.

TEAL: proposed supply-air trunks. **Dashed teal:** example room branches. **Blue dashed:** candidate outdoor-air connection at the rear facade. **EX:** separate toilet-extraction route to be coordinated.



Use the floor AHU rooms

L1-M01 and L2-M01 serve their respective floor concept. Reserve a separate ground-floor ventilation unit in the outdoor service yard. AHU counts and airflow remain provisional.

Give air a complete path

Add outdoor-air intake and supply paths. Also reserve the designed office return/extract/relief path; it is omitted from this map for clarity. Room VRF units need their own local return-air paths.

Keep toilet extraction separate

Route each toilet group to a dedicated exhaust path and suitable outdoor discharge. Do not connect toilet extract to office return air. Coordinate discharge/intake separation.

Add the VRF service route

Outdoor units -> reserved HVAC compartment -> indoor units. Model insulated refrigerant pipe sets separately from air ducts. Condensate needs its own sloping drainage route.

Check the room and crossings

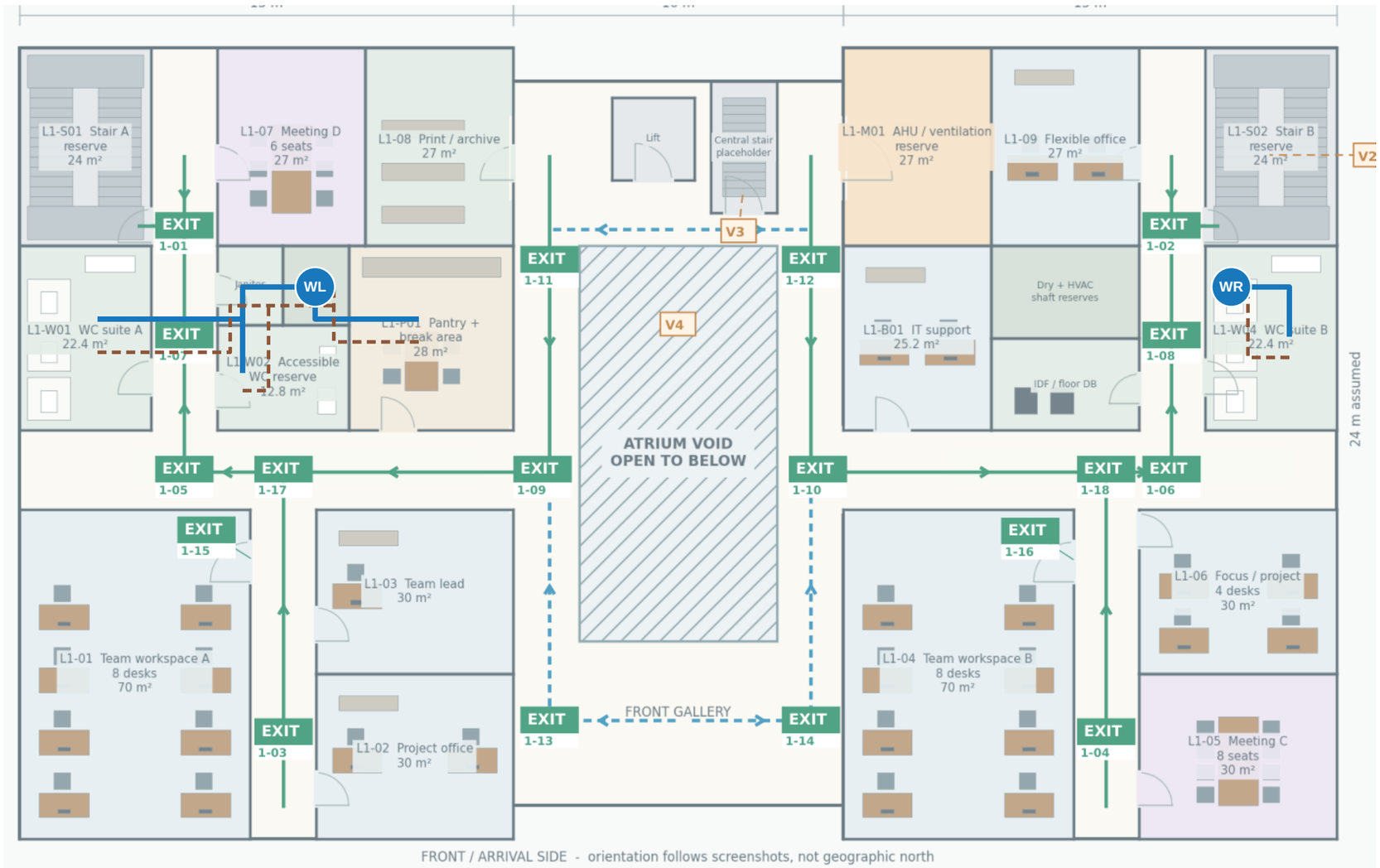
Resolve upper AHU access doors. Leave filter/coil-removal space. At rated barriers, reserve the required penetration protection, dampers and inspection access; do not assume every crossing is identical.

Model now: AHU connection faces, supply-route centerlines, separate return/extract reservations, refrigerant runs and condensate routes. The atrium smoke system remains separate: reserve possible high-level exhaust and low-level make-up air only until its strategy is engineered.

Plumbing | Keep wet services near both toilet groups

Your plan shows a left wet riser. This study also proposes a local right-side wet-service reservation to avoid a long cross-atrium drainage route.

BLUE: conceptual water connections. **BROWN DASHED:** drainage connection logic, not final slopes or elevations. **WL:** existing left reservation. **WR:** proposed new right wet-service reservation within/adjacent to the toilet zone.



Supply starts at G-M02

Domestic booster / incoming supply -> left and proposed right water risers -> local branches. Coordinate the right-side ground-floor feed through a suitable service route.

Left group

Use the existing wet-service reservation for the left toilets, accessible toilet and pantry connections. Model cold water, local hot-water provision where selected, waste and vent as separate systems.

Right group

WR is a proposed reservation, not an existing shaft. Align it through the right toilet group on all floors. Keep it separate from the adjacent electrical/IT/HVAC shaft reservations.

Wastewater goes down

Fixture drains connect to nearby stacks, then the building drain and site sewer connection. Reserve cleanouts, vent continuation and access. Drainage is not a flat ceiling distribution loop.

Resolve elevations first

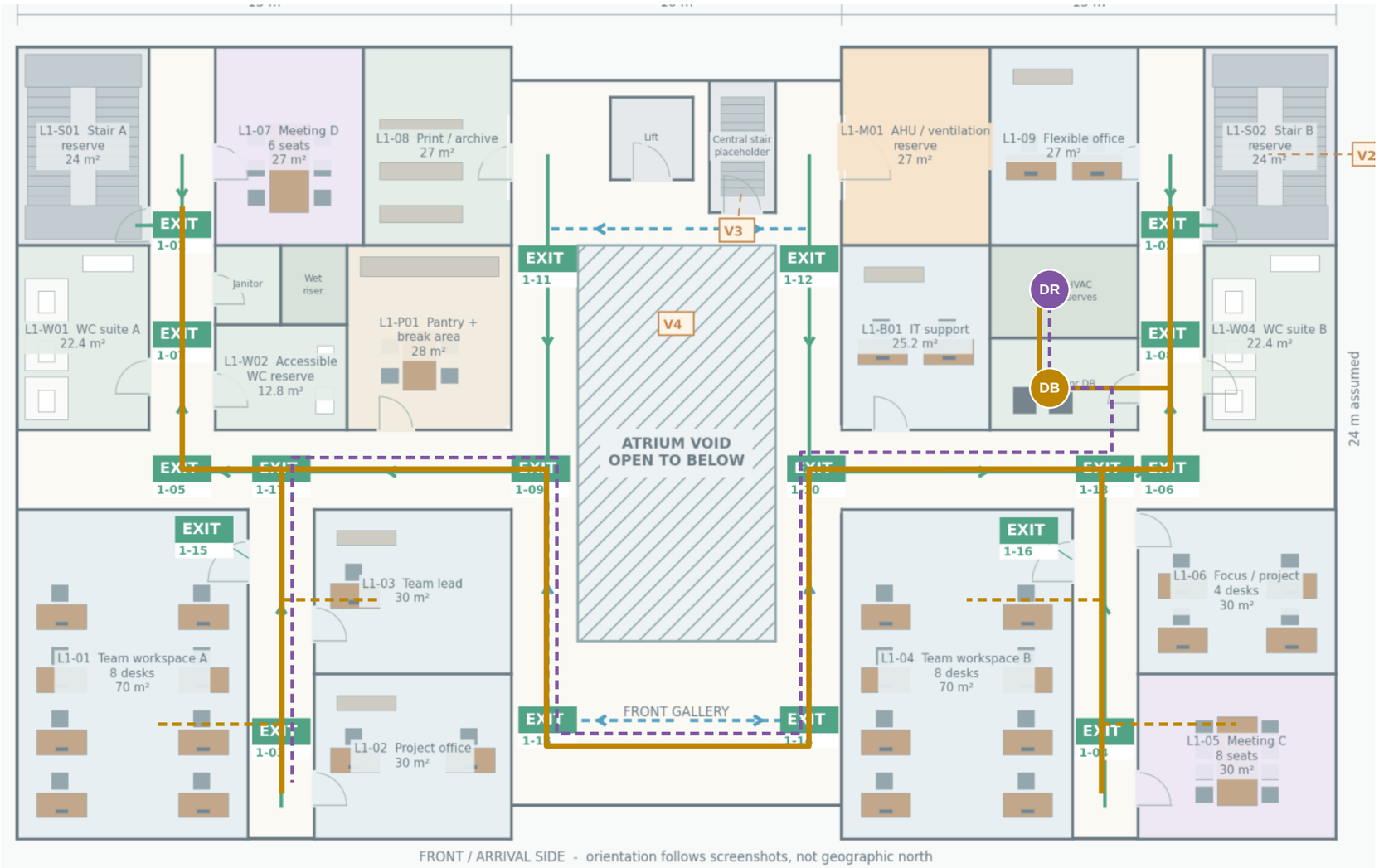
Allow continuous fall for horizontal gravity drainage and condensate. Exact slope depends on system, pipe size and adopted requirements. Use traps and approved connection details where required.

Model now: WL and proposed WR volumes; separate cold water, hot water where needed, waste and vent objects. Final shaft size, route, slope and fittings are unresolved. Keep potable water and fire-water distribution separate. Brown lines are connection intent only.

Power + data | Use separate dry pathways

The first-floor drawing illustrates corridor routing. Electrical power, data/BMS and fire-alarm circuits remain separate systems with their own sources.

GOLD: proposed power-tray route. **PURPLE DASHED:** illustrative separate data/BMS route. **DR:** dry service reservation. **DB:** floor distribution/IDF zone. Fire-alarm pathways require separate coordination and are not drawn as an IT network.



Electrical power

G-E01 main switchboard -> dedicated electrical riser compartment -> DB-G / DB-L1 / DB-L2 -> floor trays -> lighting, sockets and equipment circuits.

Network and BMS

G-I01 server/network equipment -> floor IDFs -> network devices. BMS gateways/controllers connect through their designed network and field wiring. Reserve distinct containment.

Fire alarm

The fire-alarm control panel is the circuit source, not the server rack. Reserve its dedicated field-circuit and interface pathways after panel placement and system architecture are settled.

Protect dry-service zones

Keep unrelated wet pipework away from dedicated electrical equipment space. Coordinate separation, access, supports and circuit protection. Do not treat one shared shaft as one mixed compartment.

Model in this sequence

1. Shaft volumes and ceiling zones. 2. Gravity drainage elevations and large ducts. 3. Other pipes and cable trays. 4. Branches. 5. Supports, access panels and penetration protection. 6. Room devices.

Coordination references: ICC IMC Chapter 6 (duct penetrations); ICC IPC Chapter 7 (drainage); OSHA 1910.303 (electrical dedicated space); NFPA Smoke Control Systems guidance. These are reference principles, not a determination of the locally adopted code. Keep each service on its own toggleable model layer; check insulation and access volumes as well as physical clashes.